



Clear Call

Speech Enhancement in Noisy Environments

Comparing classical DSP, deep learning, and hybrid approaches to noise suppression

ENSC 429 · DIGITAL SIGNAL PROCESSING · FINAL PROJECT · GROUP 4

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Background



Video and Voice Calls have become the default mode of professional and personal communication (Zoom, Microsoft Teams, Discord).

However, the environment people call from has changed just as much as these tools have. Remote and hybrid work means calls now happen from coffee shops, transit stations, coworking spaces, shared apartments, and even cars.

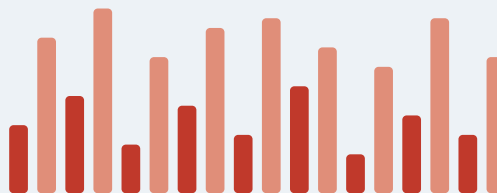
This shift led us to explore how sufficient consumer hardware is in capturing voice and rejecting acoustic clutter especially in public spaces. This then led us to speech enhancement and an opportunity to explore the ‘cocktail party’ problem.

Speech Enhancement Involves Separating a target speaker's voice from background noise

CLEAN SPEECH $s[n]$



NOISY INPUT $x[n] = s[n] + v[n]$



The algorithm only ever receives $x[n]$; $s[n]$ is used later, for evaluation only.

Problem Statement



When a user participates in a video or voice call from a public or otherwise uncontrolled acoustic environment, background noise degrades communication in several concrete ways:

- a. Ambient noise: traffic, HVAC, general environment hum
- b. Sudden transient sounds: sirens, door slams, coughing
- c. Speech-like interference: nearby voices that are difficult to distinguish from the desired speech

The goal of this project is to isolate the target's speaker's voice in real time with latency low enough

Current market solutions:

- a. Krisp: cross platform solution removes about 90% of background noise across tested noise types
- b. NVIDIA Broadcast: Hardware alternative for users with compatible GPUs. Noise cancellation is run via RTX Tensor cores and user's require NVIDIA graphic cards.
- c. Platform built in suppression

RTX Tensor Cores are dedicated hardware inside NVIDIA GPUs built specifically for neural network math, letting the noise suppression model run entirely on the GPU instead of the CPU; delivering near-zero CPU load and very low latency.

Objectives



- **Design and implement a real-time classical DSP noise-suppression pipeline** (STFT-based spectral subtraction + Wiener filtering) capable of running within a live audio with latency of about ~40ms
- **Develop and evaluate a deep-learning-based alternative** (DeepFilterNet) as a point of comparison against classical DSP approaches.
- **Explore hybrid approaches** that combine DSP and DL outputs, aiming to balance DL's audio quality against DSP's speed and predictability.
- **Establish a rigorous, quantitative evaluation methodology** - PESQ, STOI, and SNR improvement against a real benchmark dataset (VoiceBank+DEMAND) and a genuine do-nothing baseline; rather than relying on subjective listening alone.
- **Deliver a working end-to-end system** that routes enhanced audio through a virtual microphone, so it functions as a drop-in noise-suppressed input for real video-calling applications like Zoom or Teams.

Classical spectral methods → a pretrained deep-learning model → two hybrid designs that combine both.

Five Engineering Design Candidates

Each method was chosen for a specific reason and each hybrid tests a specific hypothesis

1

Original DSP

Explainable baseline: applies core course DSP concepts directly, low compute cost

2

Improved DSP

Same transparent structure, engineered to fix its weaknesses: outperformed Method 1 in dev tests

3

DeepFilterNet3

Pre-trained deep-learning comparator: used as-is, no training or fine-tuning

4

Hybrid Method 1

Fuses Improved DSP + DeepFilterNet3 waveforms to test if their strengths combine

5

Hybrid Method 2

DSP contributes only a calibration level whereas DeepFilterNet3 supplies the final waveform

Method 1: Original DSP

Frame → estimate noise → suppress → reconstruct

- 1 Framing / windowing** Divide noisy audio into short overlapping frames (Hann window)
- 2 FFT** Convert each frame to the frequency domain (real FFT, 481 bins)
- 3 Noise estimation** Minimum-statistics noise floor per bin, over a 1.2 s window
- 4 Spectral subtraction** First gain: subtract estimated noise ($\alpha = 0.3$, floor $\beta = 0.2$)
- 5 Wiener filtering** Second gain from the estimated SNR ($a = 0.7$, decision-directed)
- 6 Gain combination** Multiply both gains; apply once to the noisy spectrum
- 7 IFFT & overlap-add** Back to the time domain; 50% overlap-add reconstructs the waveform

RUNTIME MODEL

$$\mathbf{x}[n] = \mathbf{s}[n] + \mathbf{v}[n]$$

noisy = clean speech + noise; only $\mathbf{x}[n]$ is ever observed

SAMPLE RATE

48,000 Hz

FRAME / HOP

960 / 480 samples (20 / 10 ms)

NOISE WINDOW

1.2 s (120 frames)

OVER SUBTRACTION A

0.3

SPECTRAL FLOOR B

0.2

WIENER SMOOTHING A

0.7

Method 2 Improved DSP

Engineering iteration: three problems identified in Method 1, three targeted fixes

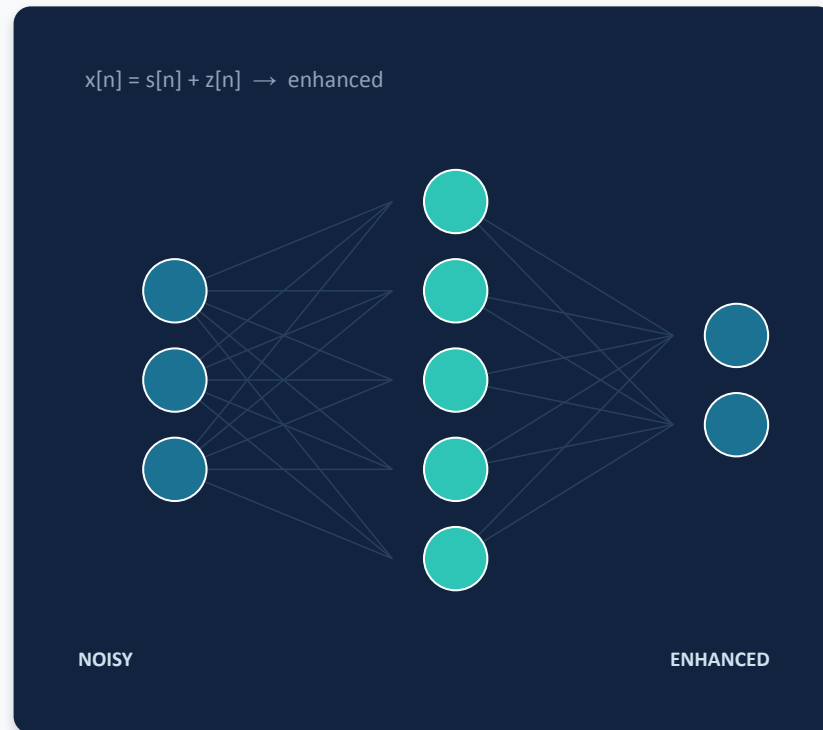
PROBLEM IDENTIFIED	ENGINEERING FIX	RATIONALE / ADDED VALUE	
Streaming-compatible framing	960-sample frame required, but live audio arrives in 480-sample blocks.	Causal adapter: $x_{\square} = [b_{\square-1}, b_{\square}]$ - combine previous + current block.	Real-time compatible; one-hop delay; keeps the 20 ms STFT geometry.
Noise estimation (MCRA)	Sliding-minimum floor reacts slowly to rising noise, with no speech check.	Speech-aware estimator: smoothing $\rightarrow$ speech-presence $p[k] \rightarrow$ controlled update.	Fast noise updates when speech is absent; frozen while speech is present.
Gain fusion & smoothing	Gains multiplied directly ($0.6 \times 0.4 = 0.24$) over-attenuate; 0.2 floor isn't final.	Soft gain fusion + a true final floor + frequency/temporal smoothing.	Recovers weak speech ($0.24 \rightarrow 0.42$); no bin drops below 20%; less musical noise.

Method 3: DeepFilterNet3

The project's pre trained deep-learning comparator

Combines classic STFT framing with a neural network that learns how to separate speech from noise. Used directly as a pretrained model with no training or fine-tuning so the project stays focused on DSP design and experimental comparison.

- 48 kHz sample rate, 20 ms window, 10 ms step
- Low bands (~0–4.8 kHz, 96 bins): multi-frame deep filter - coefficients $C[t,i,k]$ predicted by the network across 5 nearby frames
- Remaining bands: learned gain masking $G[t,k]$
- Reconstructed via inverse STFT and overlap-add



Method 4: Hybrid Method 1: Fusion

Testing the hypothesis: does combining Improved DSP + DeepFilterNet3 beat either alone?

Why: DeepFilterNet3 suppresses noise strongly but can over-attenuate weak speech (low-energy consonants); Improved DSP leaves more residual noise but preserves that weak speech better.

3.4.1 Fixed Waveform Fusion

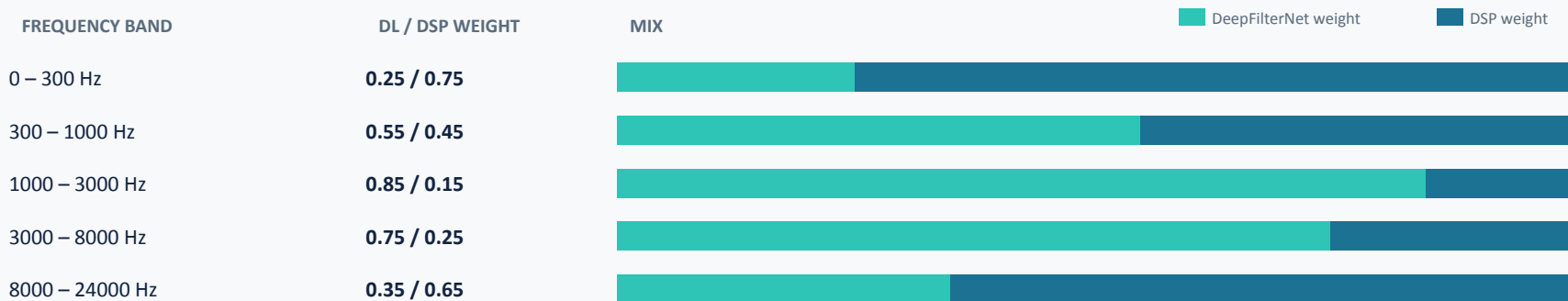
$$h[n] = \alpha \cdot l[n] + (1 - \alpha) \cdot d[n]$$

$\alpha=0.7$

3.4.2 Frequency-Band Fusion

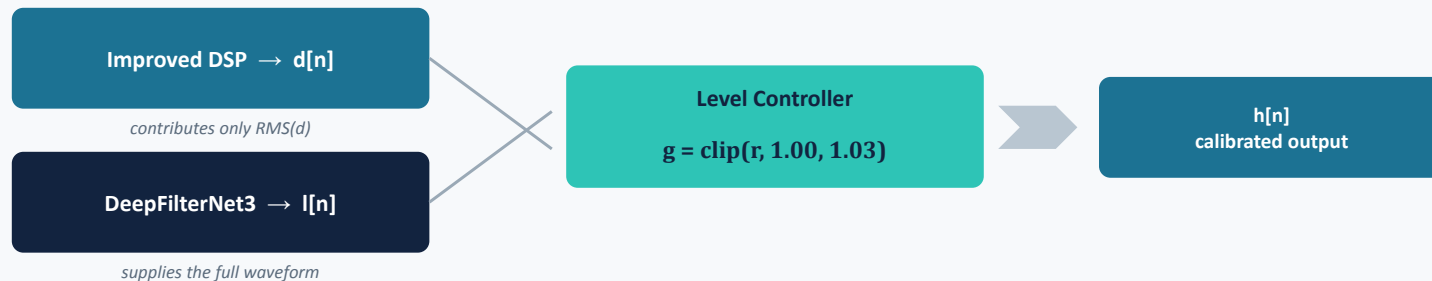
Different α per band; complex-spectrum blend on synced 20/10 ms STFTs.

Also tested: magnitude-only fusion (DSP or DL phase) to avoid phase cancellation.



Method 5: Hybrid Method 2: Level Control

A second hypothesis: give DSP a role without mixing its waveform in



- $r = \text{RMS}(d) / \text{RMS}(l)$: DSP-to-DL level ratio (RMS with $\epsilon = 1e-12$)
- Gain is bounded upward-only: 1.00 – 1.03 : never reduces the DeepFilterNet3 output, raises it by at most 3%
- Both branches near silence ($\text{RMS} < 1e-7$) → gain fixed at 1.00
- A peak-safety gain is also computed to guard against clipping

Rationale: gives Improved DSP an identifiable role in the final result, without importing its residual noise or artifacts.

Experimental Setup: Dataset & Metrics

Dataset: VoiceBank+DEMAND

~150 matched noisy/clean file pairs selected for iteration (Edinburgh DataShare).

Chosen over the DNS Challenge dataset as the right fit for course scope: matched pairs, relevant noise types, manageable size.

Pipeline config separates internal processing rate (48 kHz) from metric-computation rate (16 kHz) via *resample_for_metrics()*, preserving both real-time performance and evaluation fidelity.

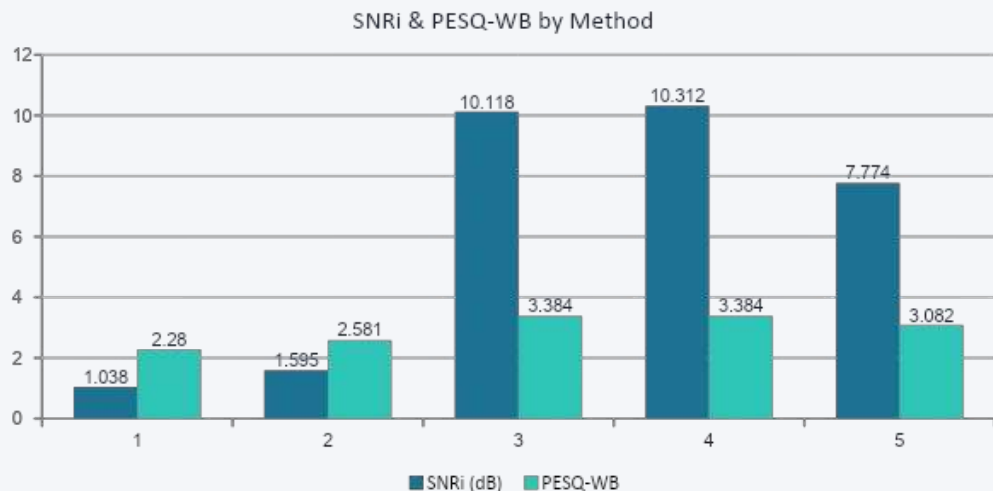
Real-Time Feasibility Test

19 noisy recordings, split into consecutive 480-sample blocks (10 ms @ 48 kHz), simulating a future streaming deployment offline.

Category	Metric
Noise Reduction	SNR Improvement (SNRI)
Speech Quality	PESQ-WB
Intelligibility	STOI
Offline Speed	Whole-clip processing-time ratio
Processing Latency	Median, maximum
Output Safety	Peak, clipping count, NaN/Inf count
Perception	Blind listening ratings

Results: Speech Quality Metrics

Evaluated on VoiceBank+DEMAND across all five methods



Method	SNRi (dB)	PESQ-WB	STOI
1. Original DSP	1.038	2.280	0.9419
2. Improved DSP	1.595	2.581	0.9469
3. DeepFilterNet3	10.118	3.384	0.9654
4. Hybrid Level Calibration	10.312	3.384	0.9654
5. Hybrid Fusion ($\alpha=0.7$)	7.774	3.082	0.9607

Headline: DeepFilterNet3 is the strongest standalone method (tied best PESQ/STOI). Hybrid Level Calibration pushes SNRi even higher without changing perceptual signal quality. Hybrid Fusion beats both DSP methods but trails DeepFilterNet3 alone.

DSP as standalone metrics perform poorly in comparison to the models using deep learning.

Results: Processing Time & Real-Time Feasibility

480-sample (10 ms) blocks at 48 kHz - offline streaming simulation, 19 recordings

Method	Median (ms)	Median Relative Cost	Max (ms)	Calls > 10 ms
Original DSP	0.0637	1.00×	0.508	0
Improved DSP	0.1116	1.75×	0.6088	0
DeepFilterNet3	0.4128	6.48×	13.8587	1
Hybrid Fusion	0.6159	9.67×	19.047	1

Reading the numbers

All methods stay well under the 10 ms per-block deadline at the median. DeepFilterNet3 and Hybrid Fusion each show one occasional slow call (max 13.9 ms / 19.0 ms) indicating a risk of occasional deadline overruns during live streaming..

Note:

These are platform-specific measurements from the laptop used for testing. Different hardware, OS conditions, background load, and power settings may shift results; they show computational feasibility, not a hardware-independent guarantee. Hybrid Level Calibration timing (0.02794 s/s ratio) is reported separately in the quality table.

Discussion: Key Findings & Tradeoffs

Original DSP

Fastest, but multiplying two attenuating gains is overly aggressive. This reduces noise but also removes weak speech.

Improved DSP

Speech-aware MCRA tracking + soft gain fusion performs better than Original DSP on SNRi, PESQ, and STOI. However it's still bounded by fixed, hand-tuned rules.

DeepFilterNet3

Best standalone method: a trained network recognizes harmonics, formants, and noise structure that fixed DSP rules cannot.

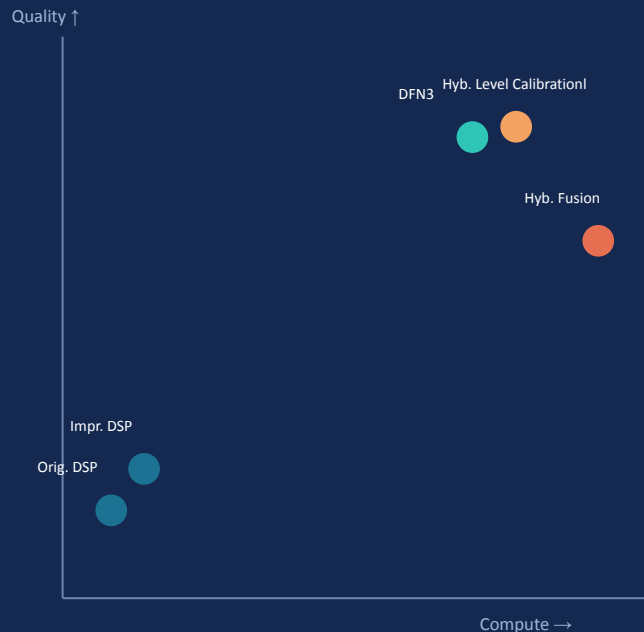
Hybrid Level Calibration

Highest SNRi via a bounded ~3% level nudge; but unchanged PESQ/STOI shows it's a calibration gain, not a denoising gain.

Hybrid Fusion

Beats both DSP methods, performing slightly better than DeepFilterNet3. Mixing in the weaker DSP waveform reintroduces residual noise. Combining two methods is not automatically better than the stronger branch.

Quality vs. Compute Complexity



Positioning is qualitative (relative SNRi/PESQ/STOI vs. median processing time), for engineering discussion — not to scale.

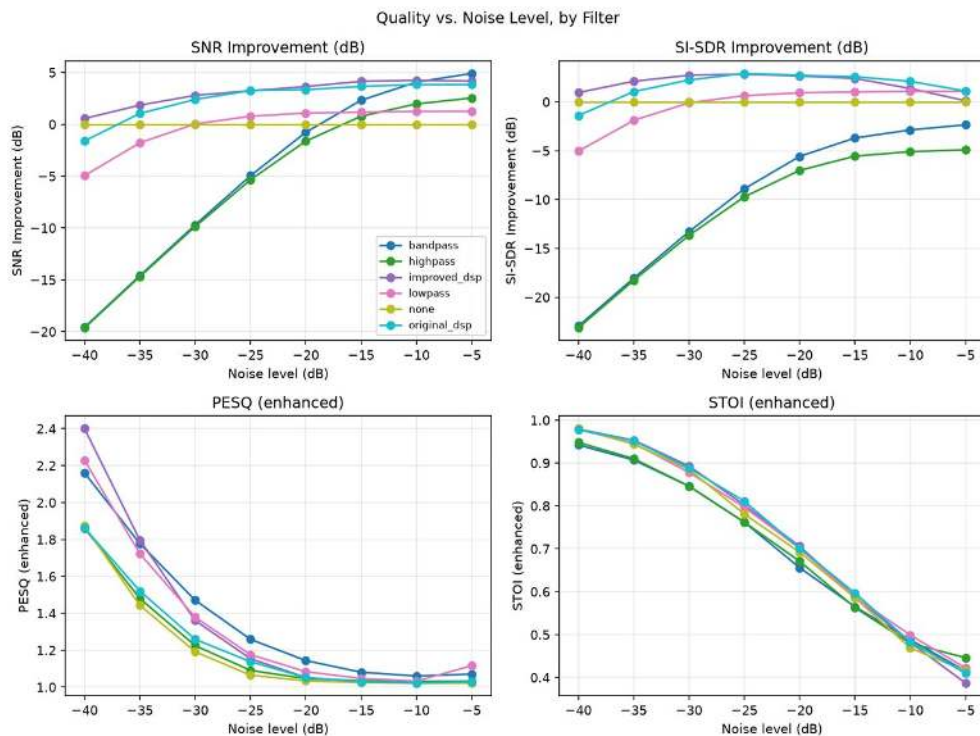
Reflection on the Proposal Plan

What changed since Week 6 and why

Proposed	Delivered	Why We Adjusted
2 methods: classical DSP vs. DeepFilterNet	5 methods: Original DSP, Improved DSP, DeepFilterNet3, Hybrid Fusion, Hybrid Level Calibration	Once the DSP baseline and DL wrapper worked, we had headroom to test real fusion hypotheses rather than stop at a single comparison.
DNS Challenge dataset	VoiceBank+DEMAND	DNS Challenge was overkill in size/scope for the course timeline; VoiceBank+DEMAND gives matched pairs and relevant noise types at a manageable size.
Cross-platform (Windows + macOS, VB-Cable/Black Hole), Zoom + Teams live test	Windows-only, VB-Cable virtual mic	Narrowing the target platform let us spend the saved integration effort on the DSP engineering iterations and the two hybrid designs instead.
Backup plan: fall back to RNNoise if classical DSP underperforms by Week 4	Not needed — Improved DSP hit dev-test targets on schedule	Original DSP's known weakness (stale noise floor) was fixed early via MCRA, so the backup trigger was never reached.

Lesson learned:

Real-Time Evaluation - Quality vs. Noise



DSP Superiority

improved_dsp and original_dsp are the only filters with consistently positive SNR/SI-SDR gains across the whole sweep.

Classic Filter Limitations

Classic bandpass/highpass filters actively hurt SI-SDR below ~-15 dB they remove noise energy but also speech structure.

Lowpass Performance

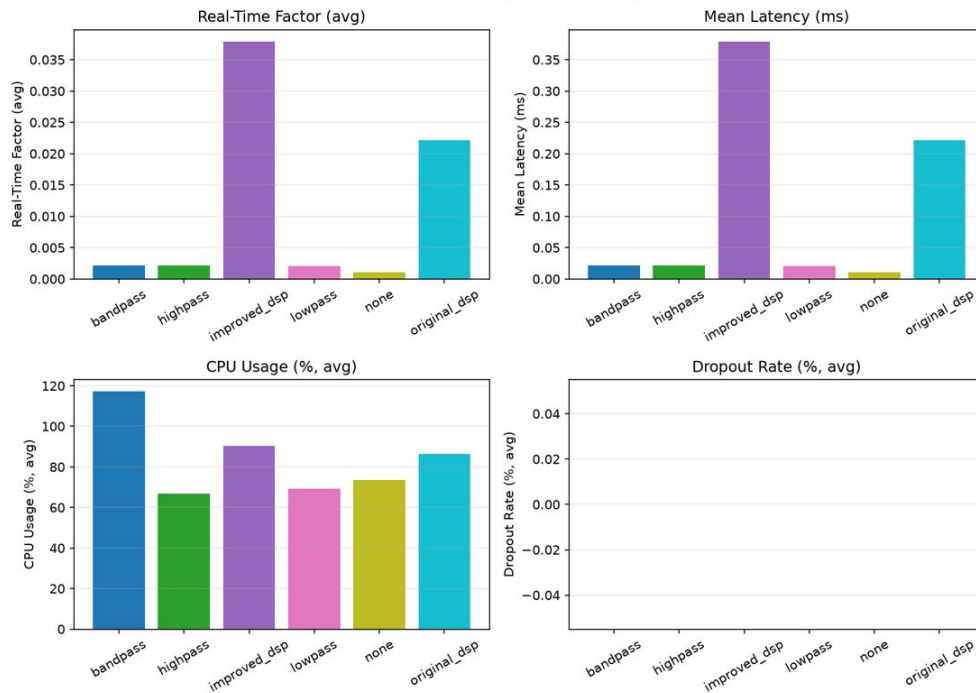
lowpass tracks closer to the DSP pipelines since speech energy is mostly low-frequency, but still trails them.

High-Noise Falloff

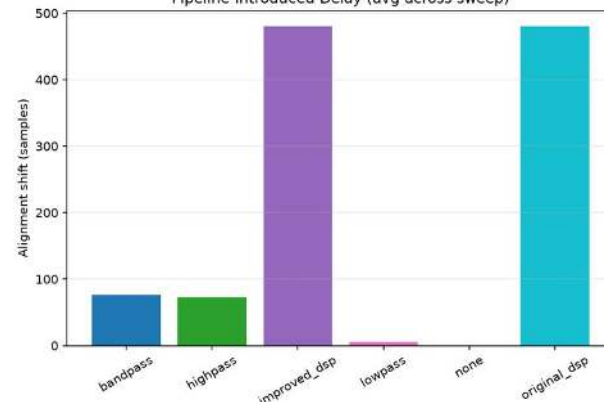
PESQ/STOI fall for every filter as noise increases, and the gap narrows at light noise (-5 dB) as there is less to enhance.

Real-Time Evaluation - Performance & Latency

Performance, Averaged Across Sweep



Pipeline-Introduced Delay (avg across sweep)



Compute Headroom & Zero Dropouts

Every filter runs at RTF well under 1.0 (worst case $\sim 0.038x$ for improved_dsp). All six have huge compute headroom and zero dropouts.

Latency & CPU Overhead

improved_dsp and original_dsp cost more CPU and carry a ~ 480 -sample (~ 10 ms) algorithmic delay from STFT frame buffering; the price of better quality.

Pre-Recorded Demo





Questions?

Speech Enhancement in Noisy Environments — ENSC 429, Group 4

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